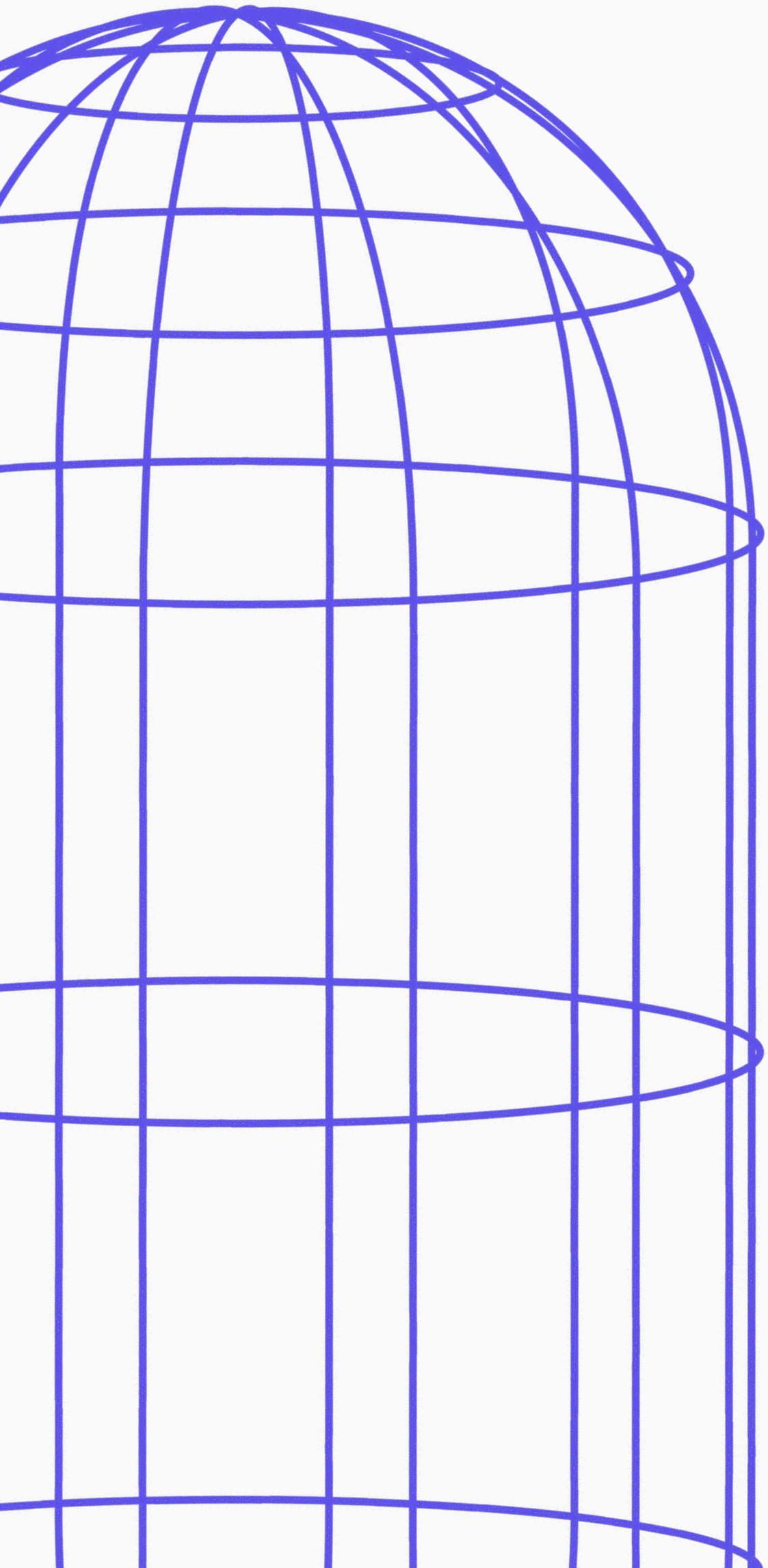
The Data Modernisation Playbook

A guide to migrating on-premises and legacy databases to Azure managed services and infrastructure



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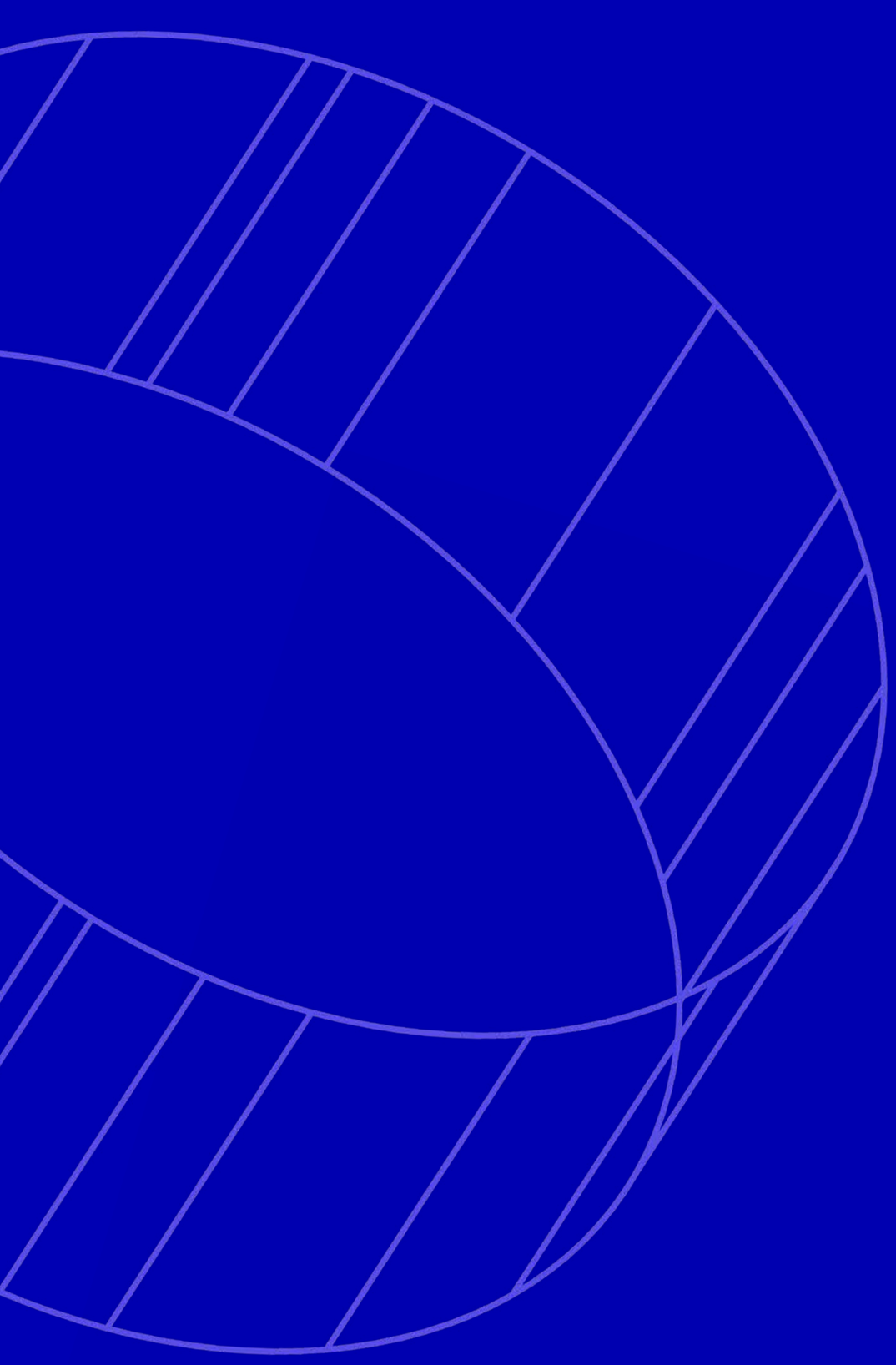
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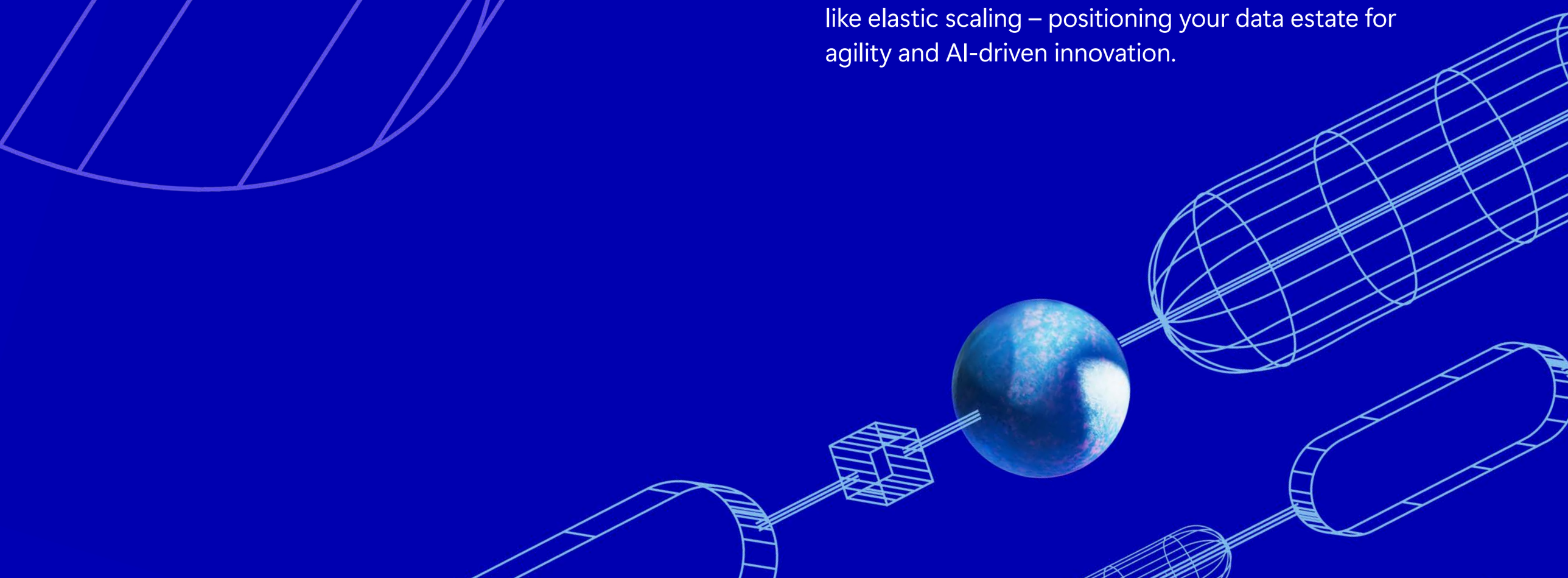
Introduction

Modernising your data is more than an infrastructure upgrade, it's a critical step to preparing for the era of agentic AI. Agentic AI is redefining how technology works, enabling AI to reason, act and collaborate across tools and data. To support this new model, organisations need a data foundation built for advanced capabilities like predictive analytics, automation, intelligent apps and autonomous agents. Yet many organisations pause modernisation efforts due to concerns about the complexity, cost, supplier lock-in or a lack of in-house skills. Meanwhile, legacy platforms, with their latency, scalability limits and compliance gaps, increasingly become barriers to innovation.

When your foundation isn't built for agentic workloads, the risks compound. First, it won't scale: latency rises, performance degrades and user experience suffers. Second, data integrity is threatened as transactions collide and processes fail under increased concurrency. Third, identity and governance models fall short – most systems were designed for human users, not autonomous agents operating at machine speed.

Addressing these concerns with a structured, risk-mitigated approach to data migration helps you build a secure, scalable and optimised foundation for current and future business needs. Database modernisation goes beyond migration by redesigning schemas, optimising indexing and adopting managed services for resilience and automation. It enables modular architectures, partitioning and cloud-native capabilities like elastic scaling – positioning your data estate for agility and AI-driven innovation.





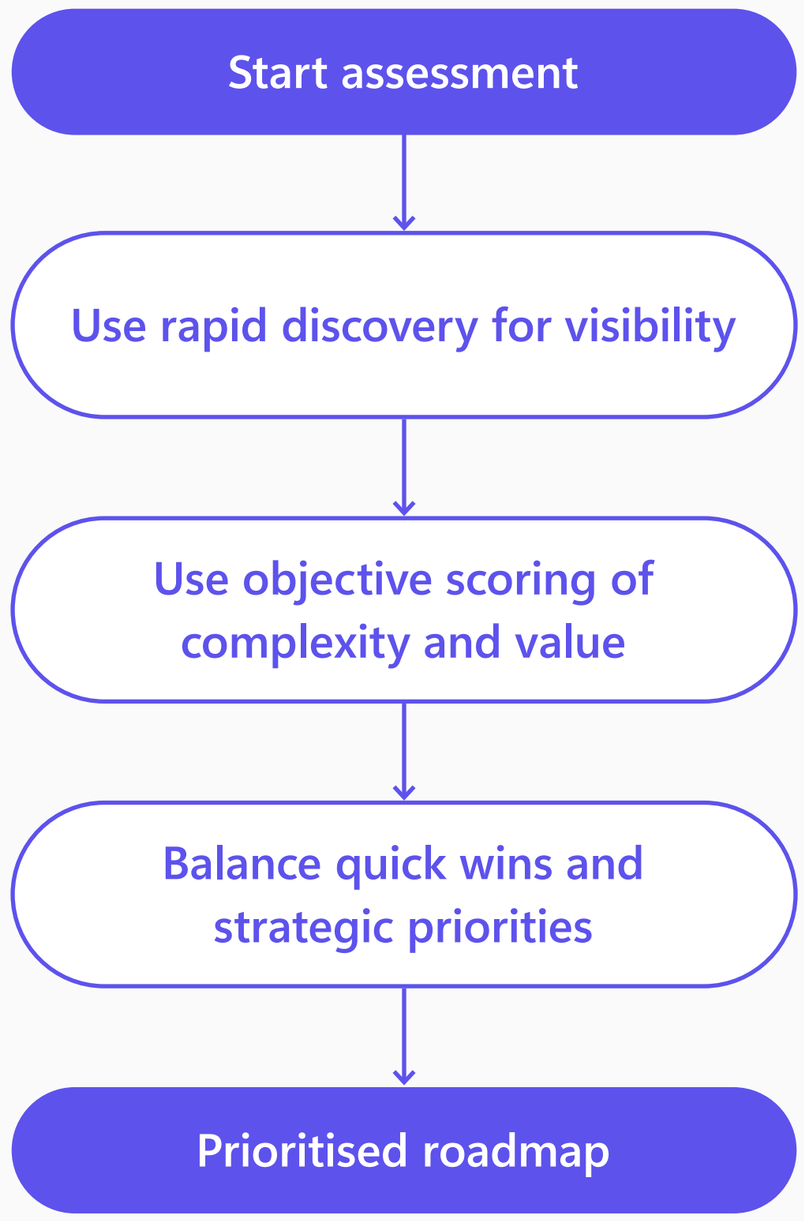
About this guide

The Data Modernisation Playbook is designed to help you unlock innovation through modernisation, equipping you to make informed decisions that improve performance, reduce costs and increase business agility. Decision-making frameworks help you choose the right path for moving on-premises and legacy database workloads to Azure Infrastructure-as-a-Service (IaaS) or Platform-as-a-Service (PaaS) options with minimal disruption.

Azure provides distinct advantages with its end-to-end approach to security, plus cost optimisation features such as Azure savings plans. With Azure's agentic- and AI-driven migration tools, you can speed modernisation efforts and reduce complexity, while robust hybrid capabilities enabled by Azure Arc provide consistent management and governance. At the infrastructure level, AMD processors and accelerators provide cost-effective computing for a wide range of workloads, helping to protect sensitive data and delivering consistent low-latency performance for users worldwide.

Frontier Accelerate for Databases is a comprehensive offering designed to simplify and speed your data modernisation journey while preparing your organisation for AI. It brings together access to Microsoft experts and specialised partners, AI-enhanced assessments and migration tooling and flexible investments – including cost savings through database savings plans, delivery funding and Azure credits – to reduce risk and accelerate outcomes. Complemented by role-based skilling and training, the programme helps teams build the expertise needed to modernise with confidence and unlock greater value from their data on Azure.

1. Assess your database estate for the era of agentic AI



In the era of agentic AI, databases must do far more than process simple transactions. They need to manage massive volumes of operational data, adapt as AI workflows evolve and support advanced use cases such as conversational memory, analytics and vector-based search. Conducting a comprehensive database assessment can help you understand how prepared you are for the agentic era and what to prioritise as you go.

Even a quick assessment can help you make sure your modernisation efforts align with business priorities, reduce risk and deliver long-term value. You can combine health checks with a comprehensive inventory, dependency mapping and workload classification to paint a picture of critical gaps, compliance risks and modernisation opportunities. This data helps you prioritise next steps and decide what to modernise when and how.

An assessment starts with an inventory. Use Azure Copilot Migration Agent and Azure Arc migration experience in SQL Server to discover all databases (SQL, NoSQL, open-source, proprietary), hosting models (on-premises, cloud, hybrid), and integration points. A high-level health check helps you identify critical gaps and opportunities.

Assess your app and data estate readiness

Identifying the current state of your apps and data is essential for planning. You can use your Azure Copilot Migration Agent to identify key code changes required before migrating to Azure. Microsoft also offers funded app and data estate assessments* with Frontier Accelerate for Databases.

* Subject to approval and availability

Quick health checklist

Run built-in assessments to benchmark performance, flag end-of-support platforms and validate security posture.

A comprehensive inventory establishes a complete view of your database estate – instances, versions, hosting models, security controls and governance. Without a full inventory, you risk overlooking critical systems or compliance gaps.

Health checklist	Action items
Performance and reliability	☐ Check that latency, throughput and uptime meet service-level agreements (SLAs). ☐ Use monitoring tools to benchmark against industry standards.
Security and compliance	☐ Ensure encryption at rest and in transit. ☐ Confirm database auditing is enabled. ☐ Validate that the database configuration and data handling comply with relevant regulations.
Lifecycle status	☐ Flag databases nearing end-of-support or running on legacy platforms.
Cost and technical debt	☐ Quantify licensing overhead. ☐ Document operational inefficiencies.

Discovery and inventory checklist

A comprehensive inventory establishes a complete view of your database estate – instances, versions, hosting models, security controls and governance. Without a full inventory, you risk overlooking critical systems or compliance gaps.

Discovery checklist	Action items
Database instances and versions	☐ Catalog all SQL, NoSQL, open-source, and proprietary systems.
Hosting models	☐ Indicate whether workloads run on premises, in the cloud or in hybrid environments.
Storage and compute footprint	☐ Capture capacity, utilisation and growth trends.
Security controls	☐ Describe authentication, encryption and access policies.
Integration points	☐ List APIs, ETL pipelines and reporting tools.
Ownership and governance	☐ Assign accountability for each asset.

Dependency mapping approach

To prevent costly surprises during migration, you need to identify the relationships between databases, applications and data flows. Azure Copilot Migration Agent helps accelerate the process with agentless and agent-based tools so you can visualise inter-server and application dependencies. For schema compatibility and feature mapping, use Azure Database Migration Service with Azure Schema Conversion Tool or Data Migration Assistant.

Dependency checklist	Action items
Schema-level mapping	☐ Identify foreign keys, views, stored procedures and triggers.
Application coupling	☐ Document which apps rely on which databases.
Data lineage	☐ Trace how data flows across systems for compliance and analytics.
Impact analysis	☐ Use visualisation tools or graph-based models to predict downstream effects of schema changes.

Workload pattern identification

Another quick way to assess candidates for migration is the workload pattern – online transactional processing (OLTP), online analytical processing (OLAP) and hybrid transactional/analytical processing (HTAP).

Workload	Azure target	Best for
OLTP High-volume, low-latency transactional systems	SQL Server on Azure Virtual Machines	Lift-and-shift from SQL Server
	Azure Database for PostgreSQL	Modernise existing PostgreSQL and Oracle workloads
	Azure Database for MySQL	Modernise existing MySQL open-source workloads
	Azure DocumentDB	MongoDB-compatible workloads
	Azure SQL Managed Instance	Modernise existing OLTP workloads
OLAP Read-intensive, large-scale queries	Microsoft Fabric	Modern, SaaS-based, unified analytics platform

Prioritisation framework

When you prioritise database candidates to migrate, you can use a simple scoring method to balance speed, impact and risk. The result is a migration candidate matrix that makes it easy to assess modernisation candidates.

For example:



Quick wins represent high business value, low complexity or risk and high readiness.



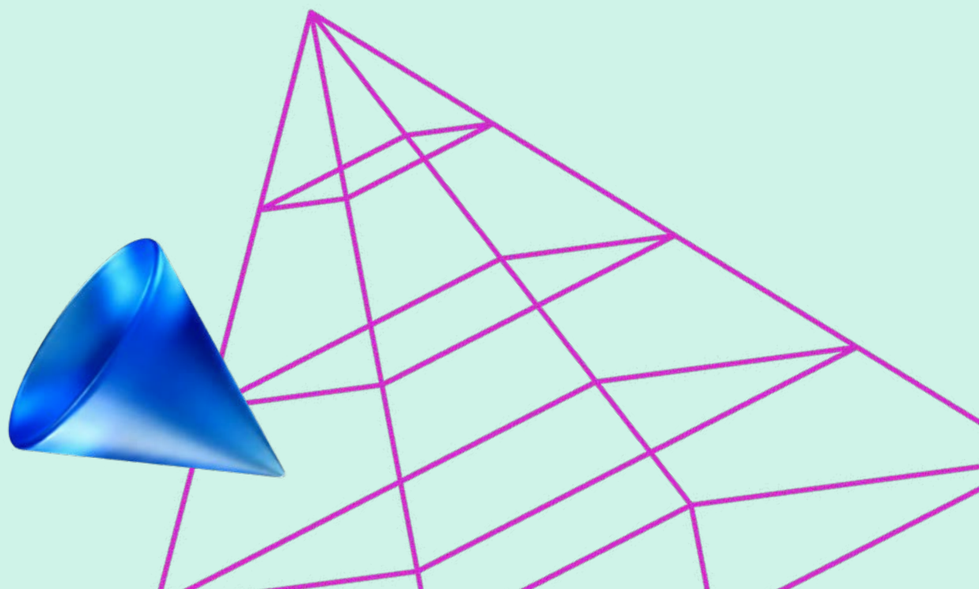
Strategic moves represent high business and technical value with greater complexity or risk and may require more planning.



Defer low-value moves when the risk, cost or complexity is too high.

Example prioritisation framework

Criteria	Database A	Database B	Database C
Business value	High	Med	Low
Technical complexity	Low	High	Med
Migration risk	Low	Med	High
Compliance requirements	Med	High	Low
Savings potential	High	Med	Low
Readiness	High	Low	Med
Strategic alignment	High	Med	Low
Candidate Type	**Quick Win**	**Strategic**	**Defer**



2. Select the right database service

Database selection is a pivotal decision point in any modernisation initiative, directly impacting performance, scalability, resilience and total cost of ownership (TCO). Azure simplifies this process by offering a broad portfolio of purpose-built database services. Combined with tools like Azure Advisor and cost-saving options such as Azure savings plans and reserved pricing, you can make informed, cost-effective decisions.

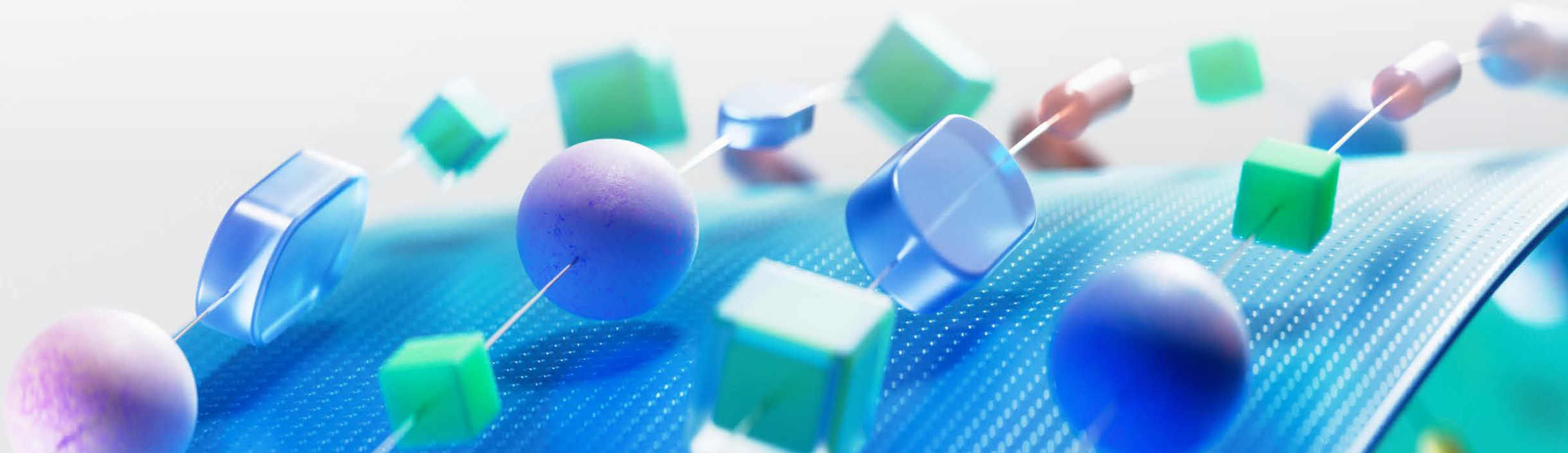
Key

1. Whether you're running multi-tenant SaaS, mission-critical OLTP or migrating on-premises SQL Server, Oracle, Sybase or IBM DB2 workloads to the cloud, Azure SQL Managed Instance offers the most frictionless path to modernisation – fully managed, always-up-to-date SQL with no infrastructure overhead.
2. Azure SQL Database Hyperscale is built for multi-tenant SaaS and high-throughput OLTP – and it's the AI-ready, hyper-scalable destination for SQL Server, Oracle, Sybase and DB2 workloads ready to modernise.
3. With full OS control, FILESTREAM, cross-instance transactions and better TCO, SQL Server on Azure VMs runs any version, any edition, on Windows or Linux.
4. Built for open-source scale, Azure Database for PostgreSQL brings enterprise security, built-in AI and mission-critical performance to Community Edition apps, GIS, timeseries and digital-marketing workloads – with a clear path from PostgreSQL and Oracle.
5. Azure Database for MySQL modernises websites, web apps and low-latency gaming on the latest Community Edition – fully managed, lower TCO and the natural destination for MySQL workloads moving to the cloud.
6. Azure DocumentDB is a fully managed, open-source, 99.03% MongoDB-compatible database. It offers built-in AI, Azure integrations and multi-cloud flexibility included for mission-critical applications.
7. Oracle AI Database@Azure delivers native Exadata performance inside Azure – full RAC, Autonomous DB, Exascale and Base DB, with the same pricing and SLAs as OCI, and with direct integration into Azure AI, analytics and security.

Selection criteria

Choosing the right database service means comparing infrastructure options, performance tiers and costs. Azure includes several tools you can use.

Assessment Tools	Data migration and schema migration	Best for
Azure Arc SQL assessment	Azure Arc migration services	Migrating SQL Server with near full SQL Server compatibility and minimal application changes.
Azure Copilot Migration Agent	Database Migration Service (DMS)	Lift and shift or compatibility dependent SQL Server workloads.
	STRIIM (online) & Database Migration Service (DMS) (offline)	Modernising to single or elastic databases with minimal downtime.
	STRIIM, SSMA for Sybase	Migrating Sybase workloads to a managed Azure SQL platform.
	STRIIM, SSMA for Oracle	Oracle-to-Azure SQL modernisation and platform consolidation.
	STRIIM, VS Code Extension: Migration Tooling	Open-source PostgreSQL migrations and cloud-native apps.
	Database Migration Service (DMS)	Straightforward PostgreSQL lift-and-shift migrations.
Azure DocumentDB Migration extension for VS Code	MongoDB-native tooling, STRIIM	MongoDB-compatible workloads.



Guide to IaaS vs. PaaS

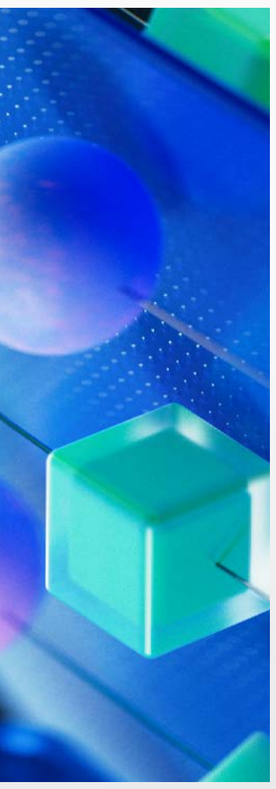
Category	IaaS	PaaS
Scaling	Manual or custom – you size and script. Useful for edge cases, regulatory needs.	Built-in autoscale and load management. Ideal for dynamic workloads and rapid growth.
Control	Full control over operating system, middleware, runtime. Best for custom configurations.	Abstracts infrastructure so you can focus on data and app logic. Less granular control.
Cost	Self-managed; can be cost-effective if tightly monitored.	Predictable costs; reduced management overhead. Pay for what you use.
Flexibility	Maximum flexibility; supports legacy and specialised workloads.	Streamlined development; best for modern, cloud-native architectures.
Management	Requires patching, back-ups, monitoring and manual scaling.	Automated patching, back-ups, scaling and high availability.
Learning curve	Familiar tools; minimal code changes.	May require new skills; enables advanced cloud features.
Compliance	Customisable for strict regulatory needs.	Built-in compliance features; may require adaptation for edge cases.
Bottom line	**You need maximum control, support for legacy workloads or have strict regulatory requirements that demand custom configurations.**	**You want faster innovation, reduced operational overhead and cloud-native scalability – ideal for most modernisation scenarios.**

Guide to database tiers

Model	Benefits	Applies To
vCore-based	Flexible, independent scaling of compute and storage.	Azure SQL Managed Instance Azure Database for PostgreSQL Azure DocumentDB Azure Database for MySQL
Serverless	Autoscales compute based on workload demand – ideal for unpredictable or intermittent workloads.	Azure SQL Database (Serverless tier)

Guide to cost optimisation

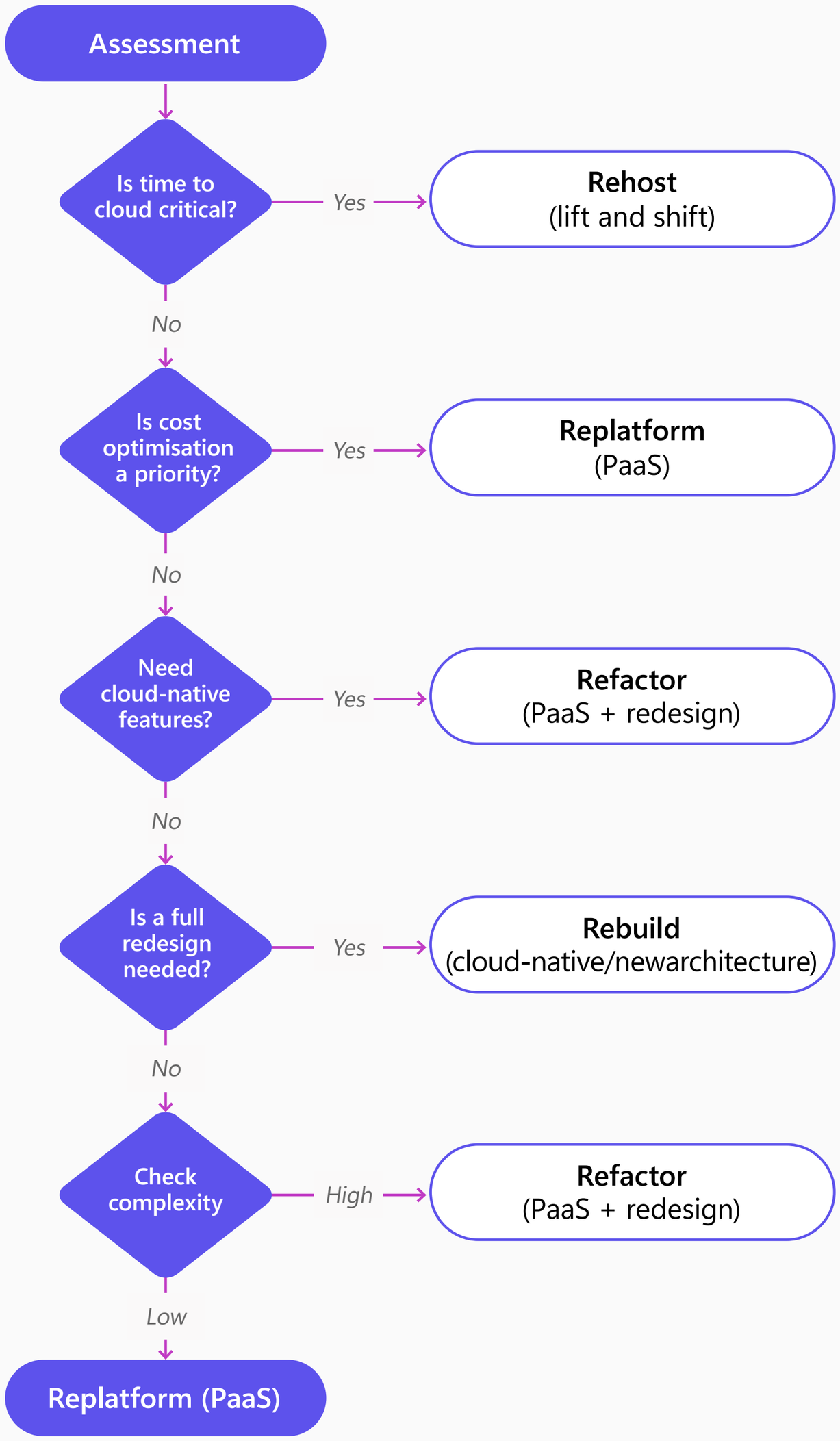
Guiding principles	Strategies and tools
Understand and forecast your costs	Estimate the cost of specific Azure services before deployment with the Azure Pricing Calculator and compare the cost of running workloads on-premises vs. in Azure.
Keep within the cost constraints	Establish spending objectives and policies using the Microsoft Cloud Adoption Framework for Azure.
Implement cost controls	Use the built-in policy template for cost management in Azure Policy, or create a custom one, so your teams can move quickly within the scope of your requirements.
Pay for consumption	Reduce costs through Azure Hybrid Benefit, Azure savings plans and Reservations that align licensing and long-term commitments with workload demand.



Tip

For SQL Server migrations, you can maximise savings by reusing existing licences through Azure Hybrid Benefit in combination with the Azure savings plan for databases, which offers spend-based savings over a one-year commitment.

3. Select the right migration pattern



There are several versions of cloud rationalisation (the 'R's), but for database modernisation, these six 'R's most often apply:

Rehost to lift-and-shift your database to VMs with minimal changes.

Replatform to migrate to managed Azure database services (PaaS) with minimal changes to schema or code.

Refactor to optimise schema, indexing or stored procedures to take advantage of cloud-native features, without a complete redesign.

Rebuild to redesign and rearchitect the database from scratch using cloud-native capabilities and modern data models.

Retire databases that are no longer needed or have been replaced.

Retain to keep as-is for now.

Selecting the right migration pattern minimises risks and maximises long-term value.

You can meet most business and technical priorities using just three common migration patterns – rehost, replatform and refactor. Whether the goal is speed, operational efficiency or cloud-native transformation, these approaches provide a clear path to modernising your data estate.

To reduce risk and accelerate decision-making, Azure Copilot Migration Agent provides visibility and Azure DMS delivers execution capability across all three paths.

Use Azure Copilot Migration Agent to:

- Plan, assess and evaluate readiness.
- Provide compute and storage estimates for IaaS and PaaS options.
- Compare cost implications across migration paths so you can make strategic decisions.

Use Azure DMS to:

- Orchestrate secure, low-downtime database migrations.
- Ensure data integrity and compliance during migration.
- Support refactoring and cross-platform conversion.
- Provide an end-to-end SQL Server path through SQL Server migration in Azure Arc.

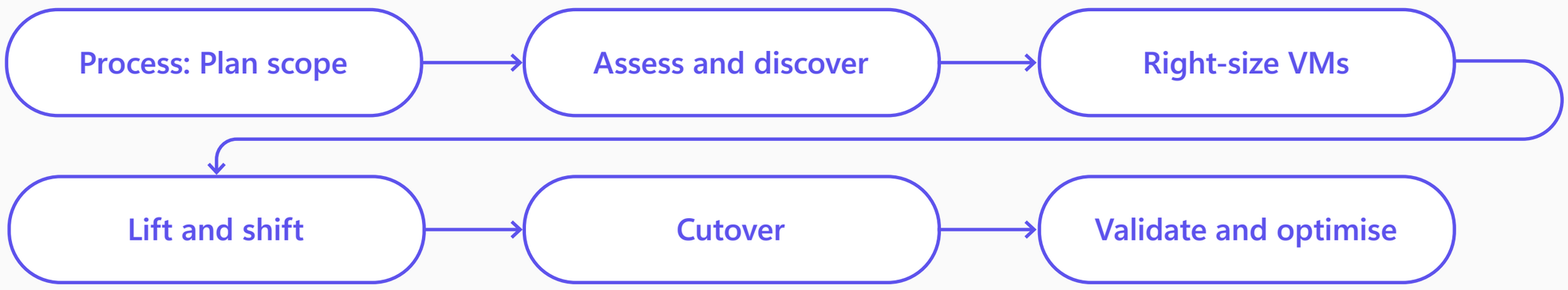


Tip

Streamline SQL Server migrations with **Azure Database Migration Service** available through Azure Arc – to complete migrations in days, not months.

Rehost: Lift-and-shift to VMs

Rehosting is a fast-track migration approach ideal for legacy workloads when you want to redeploy with minimal changes, need a quick data centre exit, face end-of-support deadlines, need to free up data centre space or simply want quick cloud ROI that reduces capital expenditures (CapEx).



Decision	Considerations
Performance and sizing	→ To avoid overprovisioning, use performance-based sizing. By taking the time to compare actual workload metrics (CPU, memory, disk I/O) from your on-premises environment, you can choose the right VM size in Azure. → To keep it simple, use as-is sizing – that is, keep the same size and specs as your current servers without changes. → Consider the processor – moving from five-year-old infrastructure to the latest v7 AMD VMs can improve performance per dollar by 78% on average across common workloads, including databases.* → After sizing, apply cost-saving options like Azure savings plan for databases. Azure Hybrid Benefit, or autoscaling where appropriate.
Downtime	→ Expect minimal to moderate downtime during cutover and plan it for a low-impact maintenance window. → Use Azure Database Migration Service to minimise downtime through continuous replication from the source to the target.
Rollback	→ Before cutover, take a snapshot or backup of VMs. → If necessary, failback to the on-premises environment using replication tools such as Azure Site Recovery.
Validation	→ Check connectivity. → Compare performance baselines. → Check backup configuration. → Assess security posture.

Expected outcomes

- ✓ Fast migration
- ✓ Better price performance
- ✓ Minimal disruption

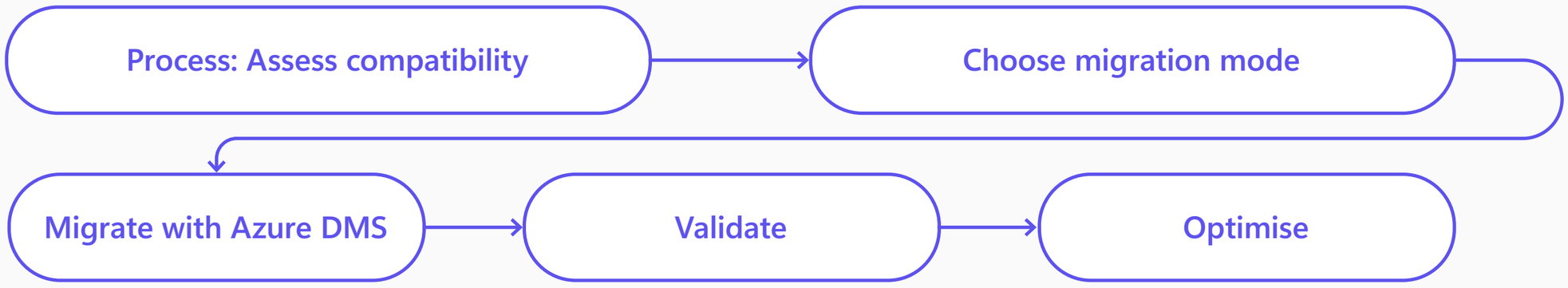
* Results may vary. See AMD EPYC™ Claim #9xx5C-104 Microsoft Azure VMs Powered by AMD EPYC™ Processors.

Tip

Rehosting is less effective when your environment is over-provisioned, apps are tied to legacy components or your organisation lacks clear modernisation goals. Consider replatforming or refactoring instead.

Replatform: Move to managed PaaS

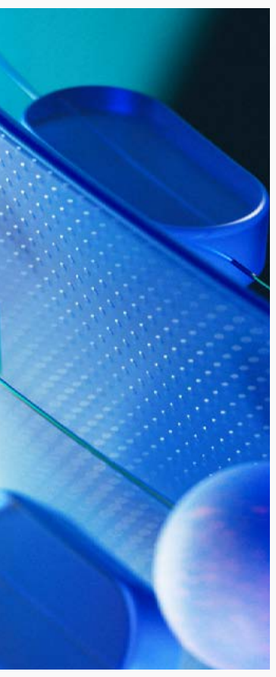
Replatforming is a modernisation approach that reduces operational overhead and unlocks scalability by transitioning to fully managed Azure database services. Replatforming is ideal when you want to modernise without significant rework. This pattern eliminates patching, back-ups and infrastructure management, while enabling elastic scaling and built-in high availability. Azure Database Migration Service streamlines the process with online and offline migration options and compatibility assessments that help you identify blockers early.



Decision	Considerations
Downtime	→ Driven primarily by data movement and cutover. → Choose a migration mode – downtime depends on online or offline mode and data volume. → Use Premium tier Azure DMS for online migrations. → Decide on a cutover strategy for switching production traffic to Azure and automate DNS changes. → Validate replication and failover readiness before switching.
Compatibility	→ Review the documentation to assess feature parity. → Ensure source data types map correctly to the target, and validate collation and character set compatibility. → Confirm authentication and security requirements. → Factor in networking and connectivity during migration, such as latency or compliance requirements. → Compare database size limits, IOPS and throughput. → Make sure the discovery process uncovers app-level dependencies like linked servers, jobs or supplier-specific functions in addition to any hard-coded database paths or platform-specific syntax.
Rollback	→ Back up source database before schema conversion. → If migration fails, restore original database and reconnect applications. → Consider creating scripts to reverse partial changes.

Decision	Considerations
Validation	→ Test functionality by verifying schema, data integrity, stored procedures and application connectivity. → Benchmark performance, comparing before-and-after migration metrics, such as query response times and CPU and memory usage. → Simulate failover to validate HA/DR mechanisms.

Expected outcomes → Significant cost reduction and performance improvement



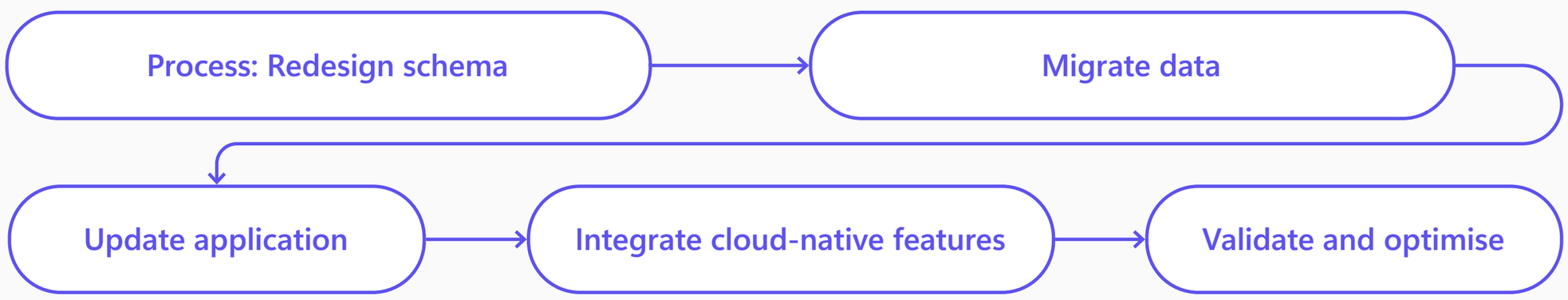
Tip

If feature incompatibility, dependency complexity or strict downtime constraints make replatforming costly or risky, consider rehosting or refactoring instead.

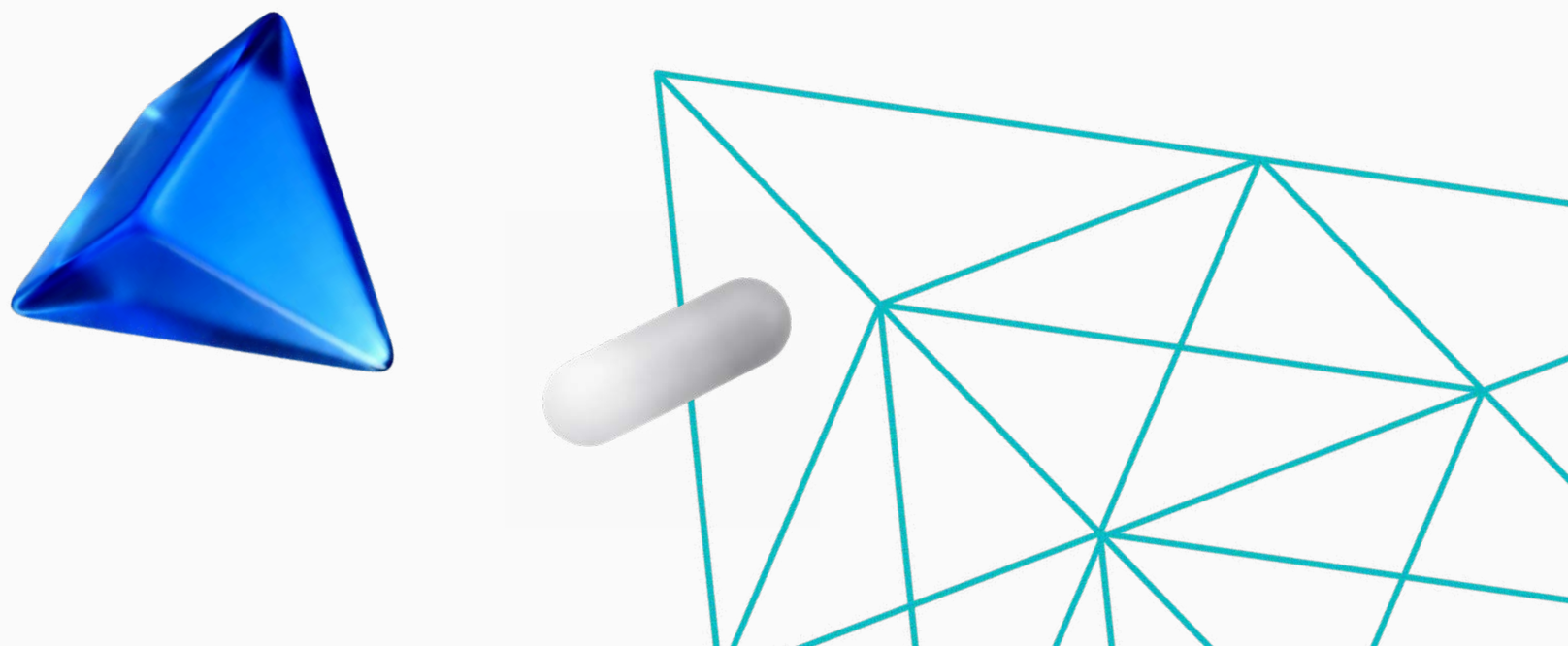
Refactor: Optimise for cloud-native

Refactoring is a transformative migration approach that unlocks advanced cloud capabilities for high-value workloads through architectural redesign. For example, you can break a monolithic schema into microservices-friendly structures, or replace hard-coded assumptions and manual failover by moving to a managed database service with dynamic resource governance and cloud elasticity.

Refactoring is ideal when your organisation is ready to embrace cloud-native design – rearchitecting databases and applications to fully integrate advanced Azure capabilities. This pattern involves optimising schemas for scalability, migrating data into modern platforms and updating application logic to integrate with cloud-native services. It's best suited for strategic workloads where performance, elasticity and innovation are top priorities.

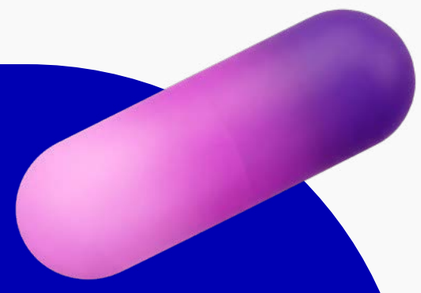


Decision	Considerations
Downtime	→ Driven primarily by applying structural changes and ensuring compatibility, which can be longer if changes are extensive. → Choose an incremental approach to reduce downtime, or use a Big-Bang approach to reduce complexity when a full cutover is okay. → Use blue-green deployment to refactor and test before switching production traffic. → Align changes with upstream and downstream systems to avoid breaking integrations during transition.
Rollback	→ Perform full back-ups plus version control for code changes. → Use phased rollout and blue-green deployments to reduce rollback risk, since application logic and database structure may have changed significantly.



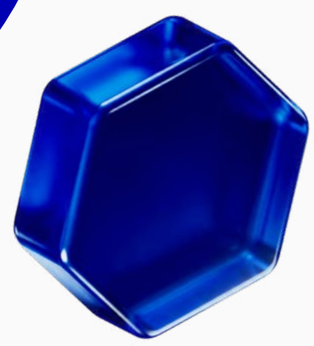
Decision	Considerations
Validation	→ Verify schema changes to make sure redesigned tables, indexes and constraints match functional requirements. → Check data integrity and transformation accuracy. → Test business logic, confirming that triggers, stored procedures and application-side logic work as intended. → Run regression tests to validate end-to-end workflows, reports and integrations after changes. → Benchmark performance, comparing query response times, concurrency handling and resource usage under load. → Validate cloud-native features, such as autoscaling, partitioning and managed identity. → Simulate HA/DR scenarios to ensure failover, backup restore and recovery processes function correctly.

Expected outcomes → Maximum performance and cost efficiency, improved agility, future-proof architecture



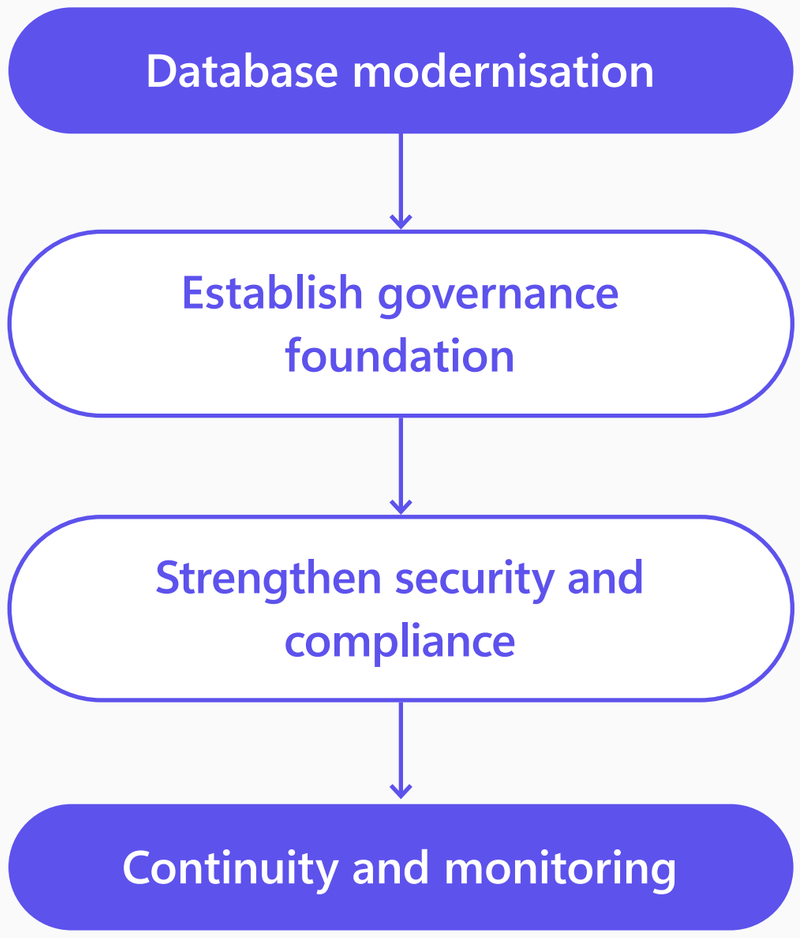
Tip

Refactoring delivers the most modernisation benefits, but demands deep planning, skilled resources and rigorous testing. If your team lacks these or timelines are tight, consider replatforming and take a phased approach to modernisation.



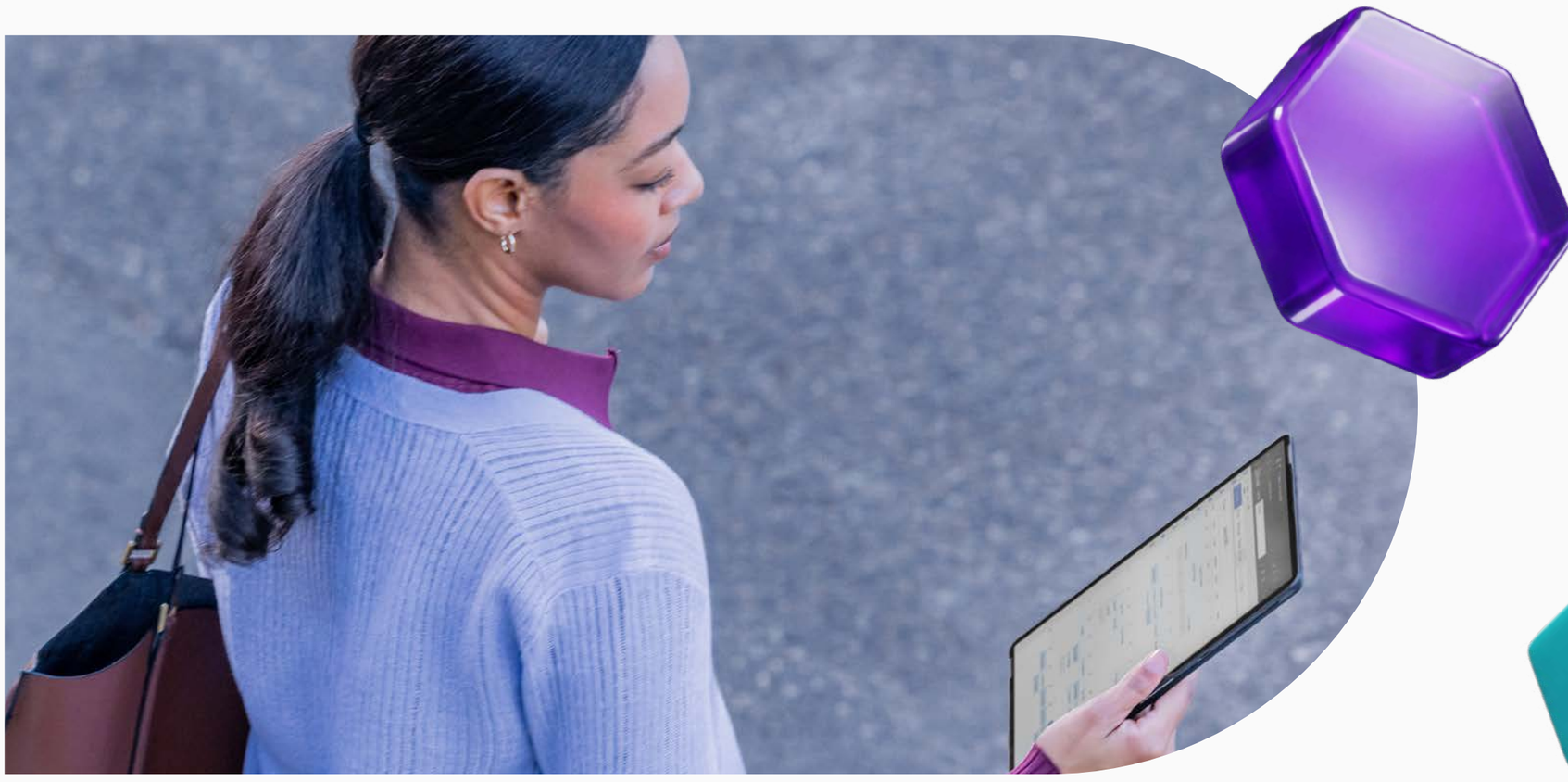
4. Implement security, integration and governance

As organisations modernise their database estate, governance becomes the backbone of a secure, scalable and AI-ready data foundation. Without it, integration complexity, security gaps and compliance risks can stall innovation and inflate costs.



Azure services work together to simplify governance and help you maintain consistent security across all deployments. You can use Microsoft Entra ID for identity and access control, while Azure Key Vault manages encryption keys and secrets. For sensitive workloads, Azure Confidential Computing provides hardware-based protection for data in use on AMD-powered infrastructure – the confidential computing platform behind Microsoft Entra ID. Microsoft Defender for Cloud is a priority for strengthening your security posture and automating threat detection. For governance and compliance, Azure Policy enforces standards and monitors resources. In hybrid or multicloud environments, Azure Arc extends these capabilities, enabling centralised policy enforcement and security management through a unified control plane.





Best practices checklists

Access and encryption

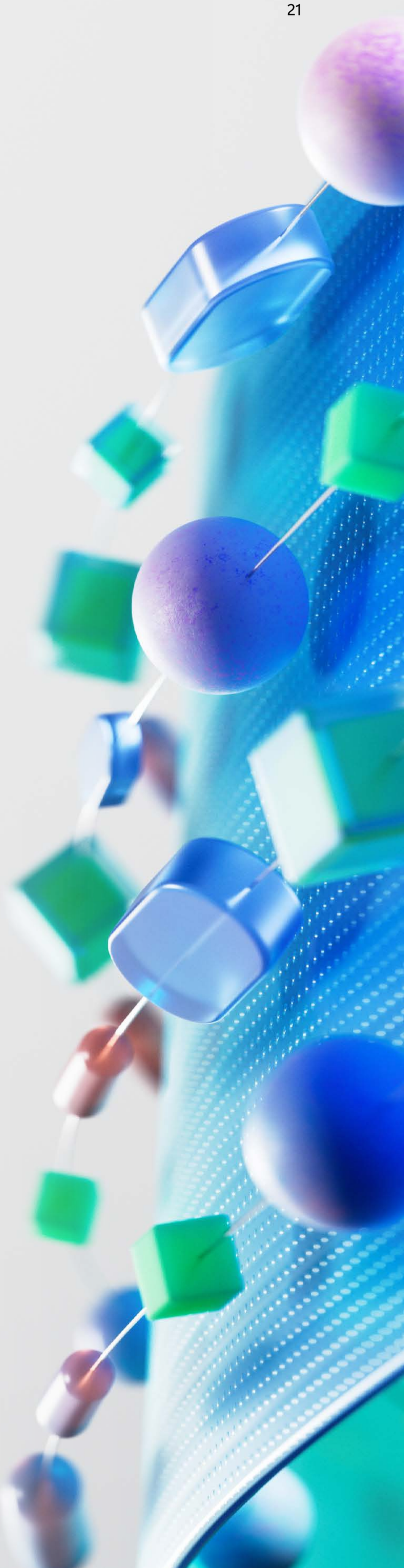
- ☐ Implement role-based access control (RBAC) and policy enforcement aligned with regulatory needs.
- ☐ Secure the connections between your modernised data and your applications – update connection strings, enforce proper authentication and configure network access for cloud and hybrid scenarios. For example, set up private endpoints and managed identities.
- ☐ Support least privilege access and encrypt and rotate keys using Azure Key Vault.
- ☐ Use Azure Confidential Computing for sensitive workloads to help protect data in use with hardware-based isolation on AMD-powered infrastructure.
- ☐ Monitor and enforce encryption policies using Defender for Cloud.

Threat protection and compliance

- ☐ Automate enforcement and vulnerability management using Defender for Cloud to monitor threats.
- ☐ Configure access control for security-specific roles using Defender for Cloud.
- ☐ Track improvements using the Defender for Cloud Secure Score to aggregate findings into a single metric and prioritise recommendations.

Continuity and monitoring

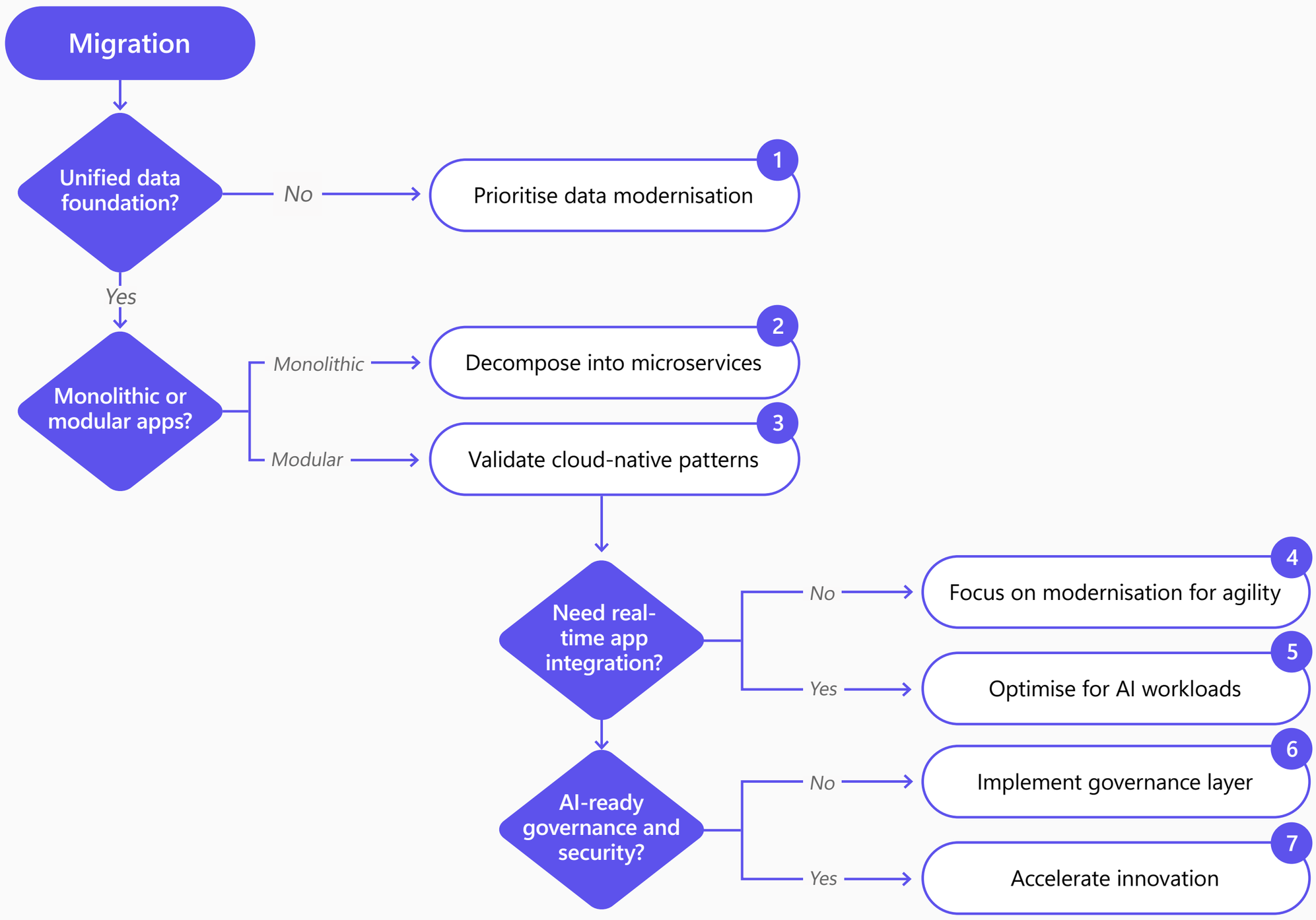
- ☐ Enable business continuity features such as automated back-ups and geo-redundant storage to meet your RPO (Recovery Point Objective) and RTO (Recovery Time Objective) targets.
- ☐ Validate backup and disaster recovery readiness using Defender for Cloud.
- ☐ Support compliance and auditing using Microsoft Purview, Audit Logs and Compliance Manager for data classification, retention and audit readiness.
- ☐ Configure monitoring and alerting using Azure Monitor, Log Analytics and alerts to track performance, availability and anomalies during and after migration.



5. Build a bridge to modernisation

Modernisation extends beyond migration by transforming your data estate into a strategic platform for advanced analytics, AI and cloud-native innovation. After moving databases to Azure, you can architect unified data lakes, enable real-time event streaming, and integrate seamlessly with Foundry Tools – capabilities that surpass traditional, siloed environments.

In addition, you can take advantage of native Azure database capabilities such as retrieval-augmented generation (RAG), real-time analytics and vector search to design intelligent applications that make the most of your own data. Data modernisation is a bridge to application modernisation, empowering your organisation to unlock competitive advantage with a unified, AI-ready data platform.



Key

1. If your data isn't unified, now is the ideal time to bring it into Azure as part of your data unification strategy. Consolidate, clean and integrate your data so it's ready for cloud-native capabilities. Implement controls for data quality, traceability and compliance early in the process. These steps prepare your organisation for advanced analytics, AI and future innovation, whether by mirroring to OneLake in Microsoft Fabric or adopting other modernisation strategies.
2. Use Azure Kubernetes Service (AKS) and containerisation to enable scalability and agility. Introduce API-first design for interoperability.
3. Ensure services are stateless where possible and use serverless for event-driven workloads.
4. Adopt managed PaaS databases and modern DevOps practices for faster iteration.
5. Integrate vector databases for RAG. Enable streaming ingestion for real-time analytics and inference.
6. Use Microsoft Purview for data cataloguing and compliance. Embed security and identity controls across data and app layers.
7. Deploy AI services (Azure OpenAI, Cognitive Services) and integrate with modern apps for intelligent experiences.

Common integration patterns

The ways in which databases interact with applications, analytics platforms and AI services is becoming increasingly sophisticated. Integration patterns provide proven blueprints for connecting modernised databases to the broader cloud ecosystem. This section highlights a few widely adopted integration patterns for Azure-based database modernisation.

AI and RAG

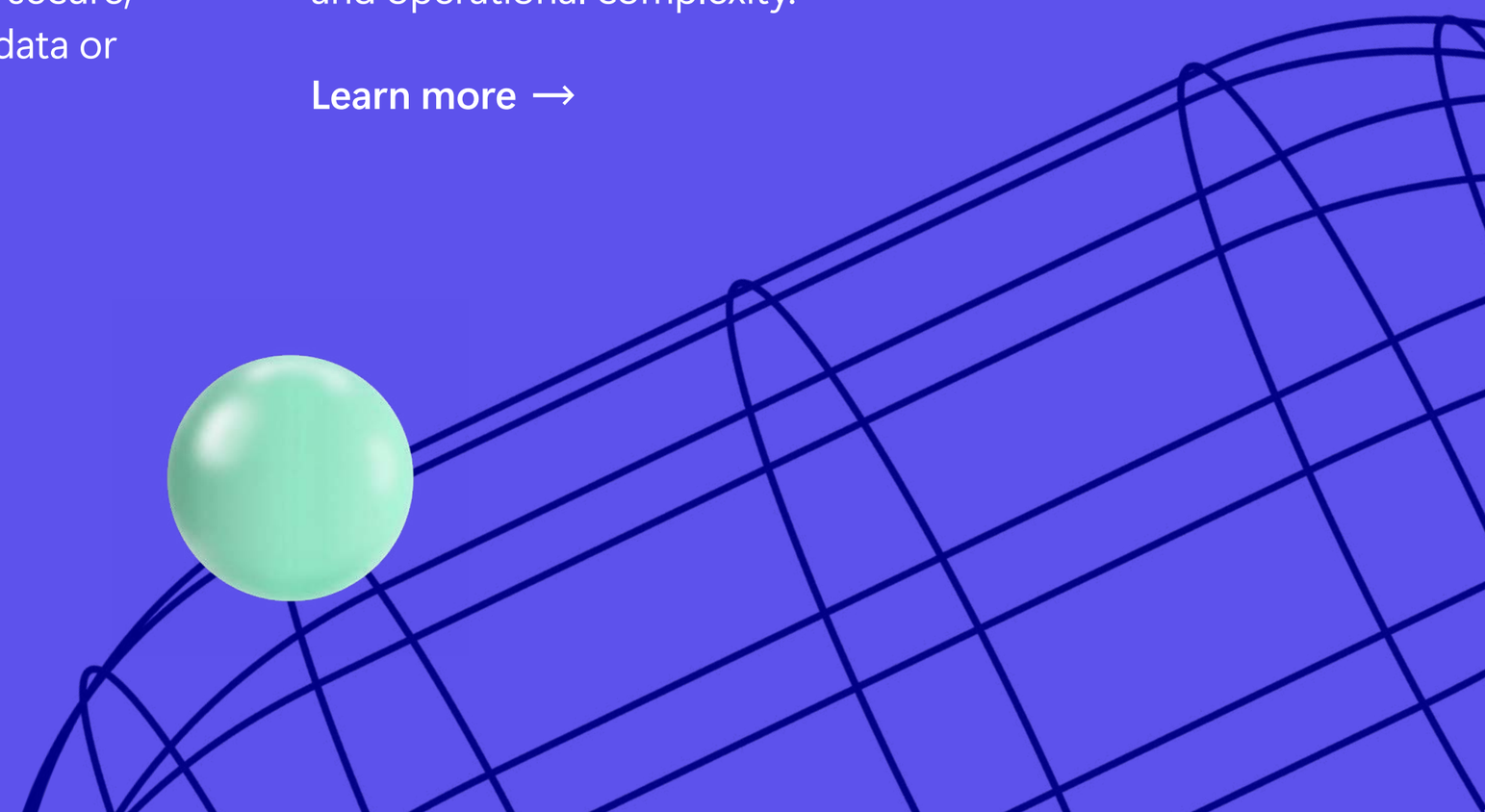
AI-driven applications need trusted, enterprise-grade data to deliver accurate and compliant responses. RAG ensures that large language models (LLMs) are grounded in authoritative sources, such as operational databases or curated data products. This pattern transforms databases from passive storage into active knowledge hubs for intelligent apps, enabling secure, context-aware insights without exposing raw data or compromising governance.

Learn more →

API-first operational access

API-first operational access ensures modern applications and microservices interact with data securely and consistently through standardised endpoints, rather than direct database connections. This approach accelerates migration to Azure, improves scalability and simplifies integration while reducing risk and operational complexity.

Learn more →



Batch and ELT ingestion

Batch and ELT ingestion remains a critical modernisation pattern. By consolidating diverse data sources into a unified, governed environment, you can create curated data products for analytics and reporting. You can optimise costs by processing large volumes efficiently while ensuring high-quality, trusted data for BI and machine learning.

Learn more →

Event-driven

From a database perspective, the event-driven pattern means capturing and reacting to data changes as they occur, rather than relying on scheduled queries or batch jobs. By streaming these events to Azure services like Change Data Capture, Azure Event Hubs or Kafka, you can provide real-time updates, asynchronous processing and robust audit trails. This approach improves agility, reduces latency and simplifies integration across modern cloud architectures.

Learn more →

Hybrid connectivity

When latency, compliance or data residency constraints prevent an immediate full cloud move, you can migrate in stages without disrupting mission-critical systems. Hybrid connectivity gives you flexibility, compliance assurance and operational continuity while you gradually adopt cloud capabilities. Azure services like ExpressRoute or VPN help ensure secure connectivity, Azure Arc provides unified management across environments, Private Link enables secure data access and Database Migration Service supports controlled migration workflows.

Learn more →

ELT analytics and zero-ETL

In modern analytics, the goal is to reduce or eliminate traditional ETL processes so you can accelerate insights without the complexity. By minimising data movement, you can enable real-time dashboards, predictive analytics and operational intelligence without waiting for batch processing cycles. For example, you can use Microsoft Fabric Real-Time Analytics to run BI and machine learning workloads on live transactional data. This approach speeds up decision-making, helps lower infrastructure costs and reduces data duplication risks, ensuring governance and compliance remain intact.

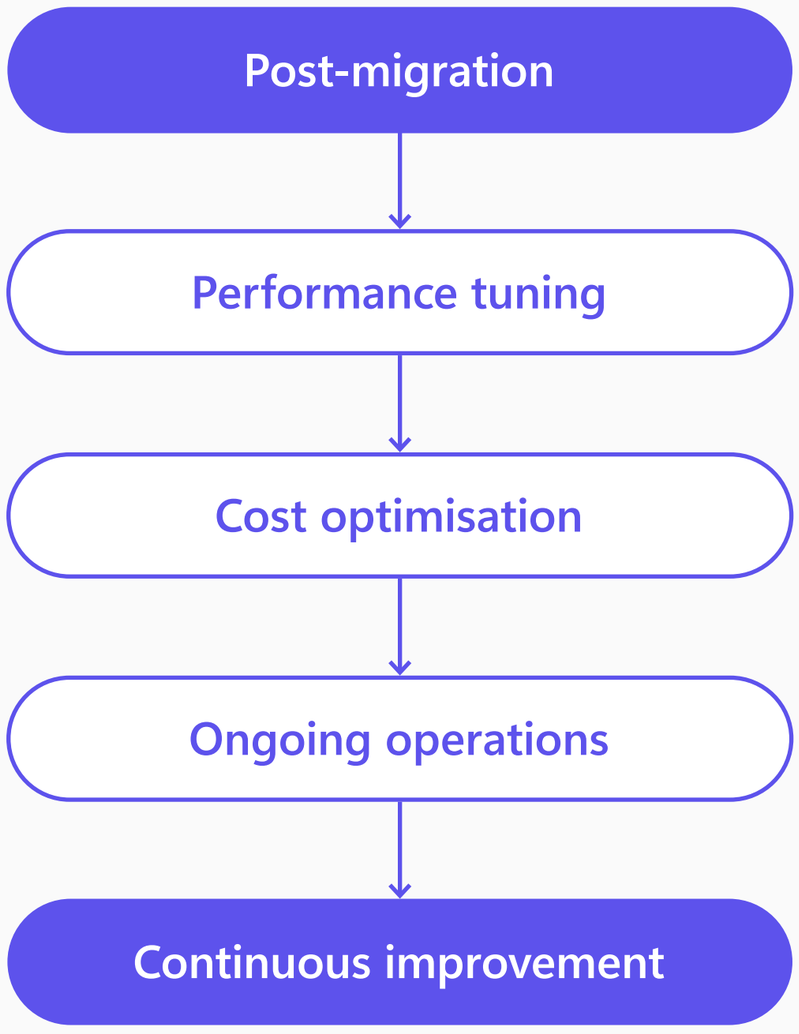
Learn more →



6. Optimise and operate

Post-migration success depends on how well your databases perform, scale and stay resilient. Without optimisation, cloud costs can spiral and performance can lag – undermining the value of modernisation. Strong governance, coupled with a culture of continuous improvement, forms the cornerstone of sustained operational excellence.

Azure provides intelligent tools to monitor, tune and optimise databases continuously. From Query Performance Insight to Cost Management, teams can make data-driven decisions that reduce overhead and improve outcomes.







Performance-tuning checklist

- ☐ Review query performance using Query Performance Insight or equivalent tools.
- ☐ Identify and apply index optimisation strategies.
- ☐ Evaluate and adjust scaling options (compute tiers, autoscale settings).
- ☐ Consider the infrastructure underneath. Azure VMs featuring AMD EPYC™ processors deliver strong performance – often at a lower cost.
- ☐ Set up monitoring and alerting using Azure Monitor and Log Analytics.
- ☐ Define maintenance windows for updates and patches.
- ☐ Schedule and validate disaster recovery tests.

Cost checklist

- ☐ Use Azure Advisor to recommend performance and cost improvements.
- ☐ Optimise workloads using design best practices in the Azure Well-Architected Framework.
- ☐ Track usage, set budgets and allocate spending with Azure Cost Management.
- ☐ Apply right-sizing recommendations, such as identifying unused or orphaned resources, increase compute density or moving workloads to PaaS or SaaS.
- ☐ Consider moving to AMD VMs to improve performance – and performance per dollar.
- ☐ Plan ahead and maximise savings by combining Azure Hybrid Benefit and Azure savings plan for databases.



Next steps

Using these practical, risk-mitigated frameworks, you can assess your data estate and choose migration patterns that balance speed, cost and long-term flexibility. By aligning governance, security and optimisation practices from the outset, you can reduce operational overhead, avoid costly rework and position your organisation for sustainable transformation.

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